

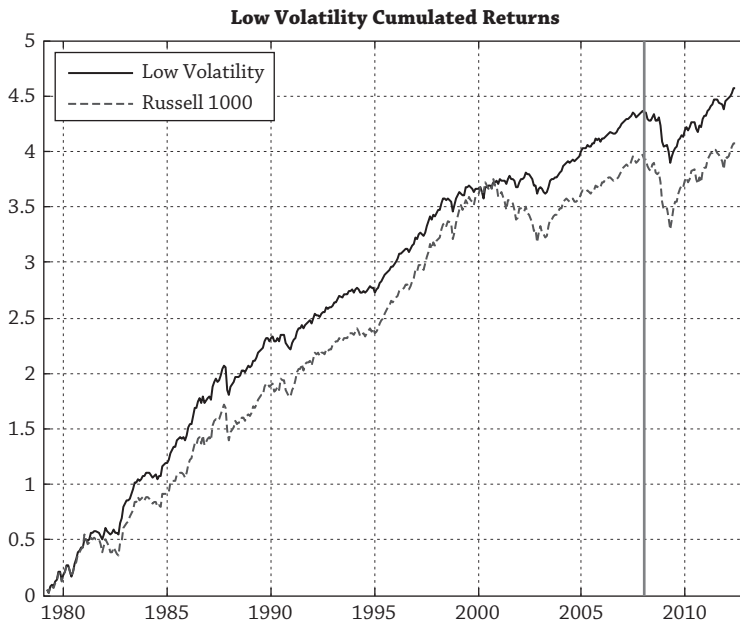


Figure 10.1

Figure 10.1 graphs cumulated returns on Martingale's Low Volatility LargeCap+ strategy based on the Russell 1000 from January 1979 to April 2012. Up until December 2007, the returns are simulated by Martingale's analysts. From January 2008, indicated in Figure 10.1 by the vertical line, performance is live.

Over the full sample, Martingale's volatility strategy beat the Russell 1000 by 1.50%. It had done even better after going live, generating an average outperformance of 1.83% since 2008. Using one-month T-bill rates as the risk-free benchmark, the low volatility strategy had a mean excess return of 8.59% and a standard deviation of 12.22%, which gave the volatility strategy a Sharpe ratio of $8.59/12.22 = 0.70$ over the full sample. In comparison, the Sharpe ratio of the Russell 1000 was 0.45. Adjusted for market risk, Martingale's performance was even more impressive; running a CAPM regression using the Russell 1000 as the market portfolio produced an alpha of 3.44% with a beta of 0.73

Scott and Herscovici, however, were not immediately swayed by these numbers and Jacques's enthusiastic presentation. Was low volatility a good fit for GM? Tracking error was closely monitored at GM Asset Management, and the Martingale product had a historic tracking error of 6.16% versus the Russell 1000. This was relatively high; other U.S. equity strategies that the group typically considered had tracking errors below 6%. Scott, as managing director of GM Asset Management's global public markets business, insisted that the equities group would need to offset that risk or convince the investment committee that a change